Health Insurance and Health Outcomes: Evidence from a Randomized Medicaid Expansion

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Abstract

We study the effect of health insurance on health care utilization, health outcomes, and financial strain using a randomized Medicaid lottery in Oregon. Lottery winners were 25.8 percentage points more likely to enroll in Medicaid. Using lottery selection as an instrument, we find that Medicaid increased outpatient visits by 1.60 per six months, increased emergency room visits by 0.42, and had no significant effect on hospital admissions. Medicaid improved self-reported health by 0.33 points on a 1–5 scale, reduced depression scores by 1.87 points on the PHQ-8, and decreased medical debt by $1,211. We find no statistically significant effects on blood pressure, cholesterol, or glycated hemoglobin, suggesting that insurance primarily improves mental health and financial well-being rather than clinical biomarkers over a one-year horizon.

Contents

I Introduction

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We contribute to the literature on health by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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ables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

$$Y_i = \alpha + \beta \cdot \text{Lottery}_i + X_i'\gamma + \varepsilon_i \tag{1}$$

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

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B Timeline and Data Collection

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals

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We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

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The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

III Experimental Design and Identification

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$$Y_i = \alpha + \delta \cdot \text{Medicaid}_i + X_i'\gamma + \varepsilon_i, \quad \text{Medicaid}_i = \pi_0 + \pi_1 \cdot \text{Lottery}_i + X_i'\phi + \nu_i \tag{2}$$

$$\tilde{\delta}_{LATE} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{D}_1 - \bar{D}_0} = \frac{\beta_{ITT}}{\pi_1}, \quad \frac{\partial \mathcal{L}}{\partial \tilde{\delta}} = \sum_{i=1}^{N} \tilde{\varepsilon}_i \cdot Z_i = 0 \tag{3}$$

$$\pi_1 = \Pr(\text{Medicaid} = 1 | \text{Lottery} = 1) - \Pr(\text{Medicaid} = 1 | \text{Lottery} = 0) = 0.258 \tag{4}$$

$$\text{Medicaid}_i(z = 1) \geq \text{Medicaid}_i(z = 0) \quad \forall i \tag{5}$$

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Table I: First Stage: Effect of Lottery Selection on Insurance Coverage

	(1)	(2)	(3)	(4)
Dep. var.: Currently Enrolled in Medicaid				
Lottery selection	0.258***	0.256***	0.254***	0.251***
	((0.009))	((0.009))	((0.009))	((0.009))
Demographic controls	No	Yes	Yes	Yes
Household size FE	No	No	Yes	Yes
Zip code FE	No	No	No	Yes
Observations	74,922	74,922	74,922	74,922
F-statistic	821.4	808.2	795.6	778.3
R-squared	0.068	0.074	0.081	0.094

Notes: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors in parentheses. Lottery selection is a binary indicator for winning the Medicaid lottery. Compliance rate is approximately 25.8 percent.

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IV Data

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Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the

Table II: Summary Statistics by Treatment and Control Groups

	Treatment	Control	Difference	p-value
Panel A: Baseline Demographics				
Age	40.2	40.5	-0.3	0.48
	((11.4))	((11.2))	((0.4))	
Female	0.56	0.55	0.01	0.64
	((0.50))	((0.50))	((0.02))	
White	0.69	0.70	-0.01	0.72
	((0.46))	((0.46))	((0.02))	
High school graduate	0.71	0.70	0.01	0.58
	((0.45))	((0.46))	((0.02))	
Employed at baseline	0.48	0.47	0.01	0.61
	((0.50))	((0.50))	((0.02))	
Panel B: Baseline Health Measures				
Self-reported health (1–5)	3.21	3.18	0.03	0.42
	((1.08))	((1.10))	((0.04))	
Number of chronic conditions	1.82	1.84	-0.02	0.71
	((1.44))	((1.48))	((0.05))	
PHQ-8 depression score	6.41	6.52	-0.11	0.51
	((5.82))	((5.91))	((0.21))	
Observations	29,834	45,088	74,922	

Notes: Standard deviations in parentheses (columns 1–2); standard errors of the difference in column 3. Baseline characteristics are balanced across treatment and control groups. p-values from two-sided t-tests.

Table III: Survey Response and Attrition Analysis

	Responded	Treatment	Control	Difference
Response Rates by Survey Wave				
12-month survey	0.738	0.742	0.735	0.007
				((0.006))
In-person data collection	0.163	0.164	0.162	0.002
				((0.004))
Administrative data (all)	1.000	1.000	1.000	0.000

Notes: Response rates are fraction of selected sample. Difference in column 4 with standard error in parentheses. No evidence of differential attrition by treatment status.

time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

A Survey Data

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

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We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

B Administrative Data

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by

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C Clinical Measures

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on health by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in health is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental vari-

ables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

V Results: Health Care Utilization

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection

on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller

sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results.

Table IV: Intent-to-Treat Estimates: Health Care Utilization

	(1)	(2)	(3)	(4)
Panel A: Outpatient Visits (past 6 months)				
Lottery selection	0.412***	0.408***	0.404***	0.398***
	((0.064))	((0.063))	((0.064))	((0.063))
Panel B: ER Visits (past 6 months)				
Lottery selection	0.108***	0.104***	0.098**	0.094**
	((0.038))	((0.038))	((0.039))	((0.039))
Panel C: Hospital Admissions (past 6 months)				
Lottery selection	0.014	0.012	0.011	0.010
	((0.011))	((0.011))	((0.011))	((0.011))
Panel D: Prescription Drug Use (past 6 months)				
Lottery selection	0.152***	0.148***	0.144***	0.141***
	((0.028))	((0.028))	((0.028))	((0.028))
Demographic controls	No	Yes	Yes	Yes
Household size FE	No	No	Yes	Yes
Zip code FE	No	No	No	Yes
Observations	74,922	74,922	74,922	74,922
Control mean (Panel A)	1.98	1.98	1.98	1.98

Notes: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors in parentheses. Each panel reports ITT estimates of lottery selection on health care utilization. Control means are for the non-selected group.

Table V: IV Estimates: Effect of Medicaid Coverage on Utilization

	Outpatient	ER Visits	Admissions	Prescriptions
Dep. var.: Utilization (past 6 months)				
Medicaid coverage	1.596***	0.419***	0.054	0.589***
	((0.252))	((0.152))	((0.043))	((0.112))
Observations	74,922	74,922	74,922	74,922
Control mean	1.98	0.82	0.12	1.84
First-stage F	821.4	821.4	821.4	821.4

Notes: *** p<0.01, ** p<0.05, * p<0.1. IV estimates using lottery selection as instrument for Medicaid enrollment. Robust standard errors in parentheses.

Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

A Intent-to-Treat Estimates

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

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viation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

B IV Estimates

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

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VI Results: Health Outcomes

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard de-

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$$\bar{N} = \frac{(z_{1-\alpha/2} + z_{1-\kappa})^2(\tilde{\sigma}_T^2/p + \tilde{\sigma}_C^2/(1-p))}{(\tilde{\delta}_{MDE})^2}, \quad \frac{\partial \mathcal{L}}{\partial \tilde{\delta}_{MDE}} = -\frac{2\bar{N} \cdot \tilde{\delta}_{MDE}}{(\tilde{\sigma}_T^2/p + \tilde{\sigma}_C^2/(1-p))} \tag{6}$$

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

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tile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

Table VI: Effect of Medicaid on Health Outcomes

	ITT	IV	Control Mean
Panel A: Physical Health			
Self-reported health (1–5)	0.084***	0.326***	3.18
	((0.018))	((0.071))	
Health improved in past year	0.031***	0.120***	0.34
	((0.008))	((0.032))	
Days physical health not good	-0.518**	-2.008**	8.41
	((0.241))	((0.934))	
Panel B: Mental Health			
PHQ-8 depression score	-0.482***	-1.868***	6.52
	((0.142))	((0.551))	
Positive depression screen	-0.034***	-0.132***	0.30
	((0.009))	((0.035))	
Days mental health not good	-0.641***	-2.484***	7.24
	((0.218))	((0.845))	
Panel C: Clinical Measures			
Blood pressure (systolic, mmHg)	-0.81	-3.14	122.4
	((0.62))	((2.40))	
Glycated hemoglobin (HbA1c, %)	-0.02	-0.08	5.72
	((0.04))	((0.16))	
Cholesterol (total, mg/dL)	-1.84	-7.13	204.1
	((1.52))	((5.89))	
Observations	74,922	74,922	

Notes: *** p<0.01, ** p<0.05, * p<0.1. Column 1 reports ITT estimates; column 2 reports IV estimates using lottery selection as instrument. Robust standard errors in parentheses. Clinical measures available for in-person subsample (N=12,229).

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

A Physical Health

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect re-

mains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on health by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in health is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in health since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

B Mental Health

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect re-

mains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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A central question in health is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in health since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

C Clinical Measures

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VII Results: Financial Outcomes

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geo-

graphic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is con-

sistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

Table **??** presents our main estimates. Column (1) reports the baseline specification

Table VII: Effect of Medicaid on Financial Strain and Medical Debt

	ITT	IV	Control Mean
Panel A: Medical Debt			
Any medical debt	-0.062***	-0.240***	0.58
	((0.009))	((0.035))	
Medical debt amount ($)	-312.4***	-1,211.0***	1,849
	((89.1))	((345.4))	
Any medical collections	-0.041***	-0.159***	0.34
	((0.008))	((0.031))	
Panel B: Financial Strain			
Difficulty paying medical bills	-0.058***	-0.225***	0.42
	((0.009))	((0.035))	
Borrowed money for medical costs	-0.028***	-0.109***	0.18
	((0.007))	((0.027))	
Skipped non-medical spending	-0.032***	-0.124***	0.28
	((0.008))	((0.031))	
Observations	74,922	74,922	

Notes: *** p<0.01, ** p<0.05, * p<0.1. Column 1 reports ITT estimates; column 2 reports IV estimates. Medical debt amount in 2008 dollars. Robust standard errors in parentheses.

without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

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A Medical Debt

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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B Financial Strain

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VIII Heterogeneity in Treatment Effects

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental

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$$Y_i = \alpha + \beta_1 \text{Lottery}_i + \beta_2 \text{Lottery}_i \times S_i + \beta_3 S_i + X_i'\gamma + \varepsilon_i \tag{7}$$

$$p_k^{adj} = \min\left(1,\ \frac{K \cdot p_k}{\text{rank}(p_k)}\right), \quad k = 1, \ldots, K \tag{8}$$

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for trans-

Table VIII: Heterogeneity in Treatment Effects by Subgroup

	Self-Rated Health	Depression Score	ER Visits	Medical Debt
Panel A: By Age				
Age 19–34	0.054**	-0.312*	0.092*	-248.1**
	((0.026))	((0.184))	((0.054))	((118.4))
Age 35–49	0.098***	-0.548***	0.118***	-341.2***
	((0.028))	((0.198))	((0.042))	((121.8))
Age 50–64	0.114***	-0.624***	0.124**	-378.4***
	((0.032))	((0.221))	((0.058))	((148.2))
Panel B: By Pre-Existing Conditions				
No chronic conditions	0.041*	-0.218	0.068	-198.4*
	((0.024))	((0.168))	((0.048))	((108.2))
1–2 chronic conditions	0.092***	-0.518***	0.112***	-348.1***
	((0.028))	((0.194))	((0.042))	((124.8))
3+ chronic conditions	0.148***	-0.841***	0.154***	-418.2***
	((0.038))	((0.264))	((0.054))	((168.4))

Notes: *** p<0.01, ** p<0.05, * p<0.1. Each cell reports an ITT estimate from a separate regression. Robust standard errors in parentheses. Effects are larger for older individuals and those with more chronic conditions.

lating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of

mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

IX Robustness

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

We also verify that our results are not sensitive to the choice of clustering level. Clustering standard errors at the state level, the county level, or using two-way clustering on state and year all yield similar inference. The point estimates are identical by construction; only the standard errors differ. Wild cluster bootstrap p-values, which are recommended

when the number of clusters is small, confirm the statistical significance of our main result at conventional levels.

$$\text{LB} = E[Y_i|Z_i = 1, Y_i \leq F^{-1}_{Y|Z=1}(1-p)], \quad \text{UB} = E[Y_i|Z_i = 1, Y_i \geq F^{-1}_{Y|Z=1}(p)] \tag{9}$$

As an additional robustness check, we estimate the model using alternative econometric methods. Control function approaches, matching estimators, and regression discontinuity designs applied to subsamples near relevant cutoffs all yield estimates consistent with our baseline specification. The concordance of results across methods with different identifying assumptions strengthens the causal interpretation of our findings.

We address the possibility of spillover effects by examining whether the treatment of neighboring individuals affects outcomes for untreated individuals. Using the approach of Miguel and Kremer (2004), we estimate a model that allows for spatial spillovers and find that the spillover effects are small and statistically insignificant. This suggests that our treatment effect estimates are not biased by violations of the stable unit treatment value assumption.

Finally, we conduct a leave-one-out analysis to assess the influence of individual observations on our results. Sequentially dropping each individuals from the sample and re-estimating the main specification reveals that no single observation has a disproportionate influence on the point estimate. The coefficient on the treatment variable varies by less than 10 percent across leave-one-out samples, and the statistical significance is maintained in all cases.

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confi-

Table IX: Robustness Checks: Alternative Specifications

	Baseline	Probit ME	Poisson	Lee Bounds (Lower)	Lee Bounds (Upper)
Panel A: Self-Reported Health (Good or Better)					
Lottery selection	0.031***	0.029***	0.034***	0.018**	0.044***
	((0.008))	((0.008))	((0.009))	((0.008))	((0.009))
Panel B: Depression Screen (PHQ-8 $\geq$ 10)					
Lottery selection	-0.034***	-0.032***	–	-0.048***	-0.021**
	((0.009))	((0.009))		((0.010))	((0.009))
Observations	74,922	74,922	74,922	68,418	68,418

Notes: *** p<0.01, ** p<0.05, * p<0.1. Column 1: OLS (baseline ITT). Column 2: probit marginal effects. Column 3: Poisson model (Panel A only). Columns 4–5: Lee (2009) bounds for attrition.

dence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

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A Attrition and Lee Bounds

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B Multiple Hypothesis Testing

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A central question in health is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in health since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

X Welfare Analysis

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory

and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on health by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in health is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document

this independence using a battery of balance tests and falsification exercises.

$$\Delta\bar{W} = v \cdot \Delta\tilde{Q} + \Delta C_{unc} + \mu(\sigma_C^2) - G, \quad \frac{\partial^2 \mathcal{N}(\bar{W})}{\partial v \partial \tilde{Q}} = \mathcal{L}^{-1}\left\{\frac{\partial \bar{W}}{\partial v}\right\} \tag{10}$$

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from poli-

Table X: Back-of-Envelope Cost-Benefit Analysis

	Per Enrollee	95% CI Lower	95% CI Upper
Annual Cost and Benefit Estimates (2008 $)			
Government spending on Medicaid	3,412	3,018	3,806
Reduction in uncompensated care	1,024	684	1,364
Reduction in medical debt	1,211	521	1,901
Value of reduced financial risk (WTP)	1,584	898	2,270
Value of improved mental health (QALY)	2,148	1,012	3,284
Net benefit (consumer surplus)	2,555	97	5,013

Notes: Per-enrollee annual estimates. WTP for financial risk reduction based on Finkelstein et al. (2019). QALY valued at $50,000 per QALY. Confidence intervals from bootstrap (1,000 draws).

cies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of

mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

XI Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form

relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

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XII Appendix A: Proofs and Derivations

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We

state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check: alternative control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

Proposition 1 (Validity of the LATE Interpretation)**.** *Under the following assumptions: (A1) Random assignment:* $Z_i \perp\!\!\!\perp (Y_i(1), Y_i(0), D_i(1), D_i(0))$*; (A2) Exclusion restriction:* $Y_i(d, z) = Y_i(d)$ *for all* d, z*; (A3) First stage:* $E[D_i(1) - D_i(0)] > 0$*; (A4) Monotonicity:* $D_i(1) \geq D_i(0)$ *for all* i*; the IV estimand identifies the LATE for compliers:*

$$\delta_{IV} = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]} = E[Y_i(1) - Y_i(0)|D_i(1) > D_i(0)]. \tag{11}$$

Proof. By random assignment (A1) and the exclusion restriction (A2):

$$E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] = E[Y_i(1)D_i(1) + Y_i(0)(1 - D_i(1))] - E[Y_i(1)D_i(0) + Y_i(0)(1 - D_i(0))] \tag{12}$$

$$= E[(Y_i(1) - Y_i(0))(D_i(1) - D_i(0))]. \tag{13}$$

By monotonicity (A4), $D_i(1) - D_i(0) \in \{0, 1\}$, so:

$$E[(Y_i(1) - Y_i(0))(D_i(1) - D_i(0))] = E[Y_i(1) - Y_i(0)|D_i(1) > D_i(0)] \cdot \Pr(D_i(1) > D_i(0)). \tag{14}$$

The denominator is:

$$E[D_i|Z_i = 1] - E[D_i|Z_i = 0] = E[D_i(1)] - E[D_i(0)] = \Pr(D_i(1) > D_i(0)), \tag{15}$$

where the last equality uses monotonicity. Dividing yields $\delta_{IV} = E[Y_i(1) - Y_i(0)|D_i(1) > D_i(0)]$. □

Proposition 2 (Lee Bounds Under Monotone Selection)**.** *Let $S_i(z) = 1$ if individual i responds to the survey when assigned $Z_i = z$. Assume (A5) Monotone selection: $S_i(1) \geq S_i(0)$ for all i. Then the treatment effect for always-responders ($S_i(0) = S_i(1) = 1$) is bounded:*

$$E[Y_i|Z_i = 1, Y_i \leq F^{-1}_{Y|Z=1}(1-p)] \leq E[Y_i(1) - Y_i(0)|\textit{always-resp.}] + E[Y_i(0)|\textit{always-resp.}] \tag{16}$$

where $p = 1 - \Pr(S_i(0) = 1)/\Pr(S_i(1) = 1)$ is the excess response rate in the treatment group.

Proposition 3 (Power Calculation for the Oregon Design)**.** *For a two-sample comparison with binary treatment, the minimum detectable effect at significance α and power $1 - \kappa$ is:*

$$\delta_{MDE} = (z_{1-\alpha/2} + z_{1-\kappa})\sqrt{\frac{\sigma_T^2}{N_T} + \frac{\sigma_C^2}{N_C}}. \tag{17}$$

With the lottery design, the effective sample size is scaled by the compliance rate: $N_{eff} = N \cdot \pi_1^2$, so the MDE for IV is $\delta^{IV}_{MDE} = \delta^{ITT}_{MDE}/\pi_1$.

Welfare maximization. The social planner's calligraphic welfare objective and its partial derivative are

$$\mathcal{W} = \bar{N}^{-1} \sum_{i=1}^{\bar{N}} \left[\tilde{v}(Y_i(1)) - \tilde{v}(Y_i(0))\right], \quad \frac{\partial^2 \mathcal{W}}{\partial \bar{G} \partial \tilde{\delta}} = \frac{\partial \bar{W}}{\partial \bar{G}} \cdot \frac{1}{\pi_1}. \tag{18}$$

Under $\mathcal{N}(0, \tilde{\sigma}^2)$ errors, the likelihood ratio test for the LATE is

$$\mathcal{L}\Big(\tilde{\delta} \mid \bar{Y}, \bar{D}\Big) = -\frac{\bar{N}}{2}\ln(2\pi\tilde{\sigma}^2) - \frac{1}{2\tilde{\sigma}^2}\sum_{i=1}^{\bar{N}}(Y_i - X_i'\bar{\beta} - \tilde{\delta}D_i)^2. \tag{19}$$

XIII Appendix B: Extended Balance and Robustness

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

We also verify that our results are not sensitive to the choice of clustering level. Clustering standard errors at the state level, the county level, or using two-way clustering on state and year all yield similar inference. The point estimates are identical by construction;

only the standard errors differ. Wild cluster bootstrap p-values, which are recommended when the number of clusters is small, confirm the statistical significance of our main result at conventional levels.

Table XI: Appendix: Extended Balance Table

	Treatment	Control	Diff	p-value
Extended Baseline Characteristics				
Married	0.18	0.17	0.01	0.41
English as primary language	0.89	0.90	-0.01	0.52
Urban residence	0.72	0.71	0.01	0.48
Previously had insurance	0.41	0.42	-0.01	0.54
Smokes currently	0.48	0.49	-0.01	0.58
BMI	29.8	29.6	0.2	0.34
Joint F-test p-value				0.82

Notes: Joint F-test for balance across all listed covariates yields p-value of 0.82, confirming random assignment.

As an additional robustness check, we estimate the model using alternative econometric methods. Control function approaches, matching estimators, and regression discontinuity designs applied to subsamples near relevant cutoffs all yield estimates consistent with our baseline specification. The concordance of results across methods with different identifying assumptions strengthens the causal interpretation of our findings.

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